Polar Coordinates & Polar Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Coordinate Conversions (Polar <-> Rectangular)**

Convert the rectangular point $(-1, -1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 1.414,\ \theta = 225°$. $r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (-1)^2} \approx 1.414$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 225°$.

2 **[Foundational]** • **Target: Converting Equations between Polar and Cartesian**

Convert the rectangular point $(-1, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 1.414,\ \theta = 135°$. $r = \sqrt{x^2 + y^2} = \sqrt{(-1)^2 + (1)^2} \approx 1.414$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 135°$.

3 **[Foundational]** • **Target: Rose Curves and Symmetry**

Convert the rectangular point $(1, -1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 1.414,\ \theta = 315°$. $r = \sqrt{x^2 + y^2} = \sqrt{(1)^2 + (-1)^2} \approx 1.414$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 315°$.

4 **[Foundational]** • **Target: Limaçons and Special Polar Curves**

Convert the rectangular point $(1, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 1.414,\ \theta = 45°$. $r = \sqrt{x^2 + y^2} = \sqrt{(1)^2 + (1)^2} \approx 1.414$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 45°$.

5 **[Foundational]** • **Target: Coordinate Conversions (Polar <-> Rectangular)**

Convert the rectangular point $(-3, 0)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 3,\ \theta = 180°$. $r = \sqrt{x^2 + y^2} = \sqrt{(-3)^2 + (0)^2} \approx 3$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 180°$.

6 **[Foundational]** • **Target: Converting Equations between Polar and Cartesian**

Convert the rectangular point $(0, -3)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 3,\ \theta = 270°$. $r = \sqrt{x^2 + y^2} = \sqrt{(0)^2 + (-3)^2} \approx 3$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 270°$.

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7 [Foundational] • Target: Rose Curves and Symmetry

Convert the rectangular point $(0, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 1,\ \theta = 90°$. $r = \sqrt{x^2 + y^2} = \sqrt{(0)^2 + (1)^2} \approx 1$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 90°$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Limaçons and Special Polar Curves

Convert the rectangular point $(0, 2)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 2,\ \theta = 90°$. $r = \sqrt{x^2 + y^2} = \sqrt{(0)^2 + (2)^2} \approx 2$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 90°$.

9 [Intermediate] • Target: Coordinate Conversions (Polar <-> Rectangular)

Convert the rectangular point $(2, 0)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 2,\ \theta = 0°$. $r = \sqrt{x^2 + y^2} = \sqrt{(2)^2 + (0)^2} \approx 2$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 0°$.

10 [Intermediate] • Target: Converting Equations between Polar and Cartesian

Convert the rectangular point $(1, 0)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

Final Answer: $r = 1,\ \theta = 0°$. $r = \sqrt{x^2 + y^2} = \sqrt{(1)^2 + (0)^2} \approx 1$, $\theta = \tan^{-1}(y/x)$ adjusted for quadrant $\approx 0°$.

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11 **[Intermediate]** • **Target: Rose Curves and Symmetry**

Consider the function $f(x) = x^3 - 6x^2 + 9x + 1$. A line tangent to the graph of $f(x)$ at $x = 1$ intersects the graph at another point P. Determine the x-coordinate of point P.

Final Answer: 4. First, find the derivative of $f(x)$ to determine the slope of the tangent line: $f'(x) = 3x^2 - 12x + 9$. The slope of the tangent line at $x = 1$ is $f'(1) = 3(1)^2 - 12(1) + 9 = 3 - 12 + 9 = 0$. The y-coordinate of the point of tangency is $f(1) = (1)^3 - 6(1)^2 + 9(1) + 1 = 1 - 6 + 9 + 1 = 5$. The equation of the tangent line is $y - y_1 = m(x - x_1)$, which becomes $y - 5 = 0(x - 1)$, so $y = 5$. To find the intersection points, set $f(x)$ equal to the tangent line's equation: $x^3 - 6x^2 + 9x + 1 = 5$. Rearrange the equation: $x^3 - 6x^2 + 9x - 4 = 0$. We know that $x = 1$ is a point of tangency, which means it is a repeated root. Therefore, $(x - 1)^2$ must be a factor of $x^3 - 6x^2 + 9x - 4$. Expanding $(x - 1)^2 = x^2 - 2x + 1$. We can use polynomial division or synthetic division to find the other factor. Alternatively, we can reason that if $x = 1$ is a repeated root, the sum of the roots must be related to the coefficient of x^2. Let the roots be $1, 1, r$. According to Vieta's formulas, the sum of the roots is $-(-6)/1 = 6$. So, $1 + 1 + r = 6$, which gives $r = 4$. Thus, the other intersection point has an x-coordinate of 4.

12 **[Intermediate]** • **Target: Limaçons and Special Polar Curves**

A right-angled triangle ABC is shown with the right angle at vertex C. The length of side AC is 8 units and the length of side BC is 6 units. What is the value of $\sin(\angle BAC)$? Show your complete algebraic work.

Final Answer: $\frac{3}{5}$. The triangle ABC is a right-angled triangle with the right angle at C. We are given AC = 8 and BC = 6. We want to find $\sin(\angle BAC)$. In a right-angled triangle, the sine of an angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. First, we need to find the length of the hypotenuse AB using the Pythagorean theorem: $AB^2 = AC^2 + BC^2$. $AB^2 = 8^2 + 6^2$ $AB^2 = 64 + 36$ $AB^2 = 100$ $AB = \sqrt{100} = 10$ units. Now, we can find $\sin(\angle BAC)$: $\sin(\angle BAC) = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{BC}{AB}$ $\sin(\angle BAC) = \frac{6}{10} = \frac{3}{5}$. Therefore, the value of $\sin(\angle BAC)$ is $\frac{3}{5}$.

13 **[Intermediate]** • **Target: Coordinate Conversions (Polar <-> Rectangular)**

The graph of the quadratic function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at two points. What is the y-coordinate of the vertex of this parabola? Show your complete algebraic work.

Final Answer: -1. The vertex of a parabola in the form $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. For the function $f(x) = x^2 - 4x + 3$, we have $a = 1$ and $b = -4$. Thus, the x-coordinate of the vertex is $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, we substitute this value of x back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the y-coordinate of the vertex is -1.

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14 [Intermediate] • Target: Converting Equations between Polar and Cartesian

The graph of the exponential function $f(x) = 2^x$ is given. A point P on the graph has a y-coordinate of 8. What is the x-coordinate of point P? Show your complete algebraic work.

Final Answer: 3. The function is given by $f(x) = 2^x$. We are looking for a point P on the graph where the y-coordinate is 8. This means we need to solve for x when $f(x) = 8$. So, we have the equation $2^x = 8$. Since $8 = 2^3$, we can rewrite the equation as $2^x = 2^3$. By equating the exponents, we find that $x = 3$. Therefore, the x-coordinate of point P is 3.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Rose Curves and Symmetry • Cross-Lesson: 1.03 Rates of Change & Concavity

$\tan\theta = \frac{\text{opp}}{\text{adj}}$. In 3-4-5 triangle with opp=3,adj=4:.

Final Answer: $\frac{3}{4}$. Applying Unit 3 · Trigonometric and Polar Functions principles, $\tan = \frac{3}{4}$ Therefore, the verified correct option is $\frac{3}{4}$.

16 [Challenging] • Target: Limaçons and Special Polar Curves • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Period of $y = \sin(2x)$ is:.

Final Answer: π. Period $= \frac{2\pi}{b} = \frac{2\pi}{2} = \pi$

17 [Challenging] • Target: Coordinate Conversions (Polar <-> Rectangular) • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Consider a right-angled triangle ABC, where the right angle is at vertex B. Point D lies on the hypotenuse AC such that BD is perpendicular to AC. If AB = 10 units and BC = 24 units, find the length of AD.

Final Answer: $\frac{100}{26}$. Let the triangle be ABC with $\angle B = 90^\circ$. $BD \perp AC$, with D on AC. By the Pythagorean theorem in ΔABC, we have $AC^2 = AB^2 + BC^2$. Given $AB = 10$ and $BC = 24$, we get $AC^2 = 10^2 + 24^2 = 100 + 576 = 676$. So, $AC = \sqrt{676} = 26$ units. In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. $\Delta ABC \sim \Delta ADB \sim \Delta BDC$. From the similarity $\Delta ABC \sim \Delta ADB$, we have the ratio of corresponding sides: $\frac{AB}{AC} = \frac{AD}{AB}$ Substituting the known values: $\frac{10}{26} = \frac{AD}{10}$ Solving for AD: $AD = \frac{10 \times 10}{26} = \frac{100}{26}$ units. Therefore, the length of AD is $\frac{100}{26}$. Common traps: - Confusing the segments of the hypotenuse (AD and DC). - Using incorrect similarity ratios. - Calculation errors with squares and square roots.

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18 [Challenging] • Target: Converting Equations between Polar and Cartesian • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Consider the exponential function $f(x) = a \cdot b^x$ where a and b are positive constants. The graph of $y = f(x)$ passes through the points $P = (1, 6)$ and $Q = (3, 54)$. A related function $g(x)$ is defined such that $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. If the point R on the graph of $y = g(x)$ has an x-coordinate of 2, what is the y-coordinate of R? Show your complete algebraic work.

Final Answer: 2. The function is given by $f(x) = a \cdot b^x$. It passes through $(1, 6)$ and $(3, 54)$. \\ We have two equations: \\ 1) $6 = a \cdot b^1$ \\ 2) x \\ $9 = b^2$ \\ Since b is a positive constant, $b = 3$. \\ Substitute $b = 3$ into equation (1): \\ $6 = a \cdot 3$ \\ $a = 2$ \\ So, the function is $f(x) = 2 \cdot 3^x$. \\ The related function is $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. Substituting $a = 2$ and $b = 3$, we get: \\ $g(x) = \log_3\left(\frac{2 \cdot 3^x}{2}\right)$ \\ $g(x) = \log_3(3^x)$ \\ Using the logarithm property $\log_b(b^x) = x$, we have: \\ $g(x) = x$ \\ The point R on the graph of $y = g(x)$ has an x-coordinate of 2. Therefore, the y-coordinate of R is $g(2) = 2$. \\ The correct choice is 2.

19 [Challenging] • Target: Rose Curves and Symmetry • Cross-Lesson: 4.01 Inverse & Composite Functions

The terminal ray of an angle θ in standard position in the xy-plane passes through the point $(-5, 12)$. What is the value of $\sin\theta$? Show your complete algebraic work.

Final Answer: $\frac{12}{13}$. For an angle θ in standard position whose terminal ray passes through a point (x, y), the trigonometric functions are defined using the distance from the origin to the point, $r = \sqrt{x^2 + y^2}$. 1. Calculate r: $r = \sqrt{(-5)^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$ 2. Compute $\sin\theta$: $\sin\theta = \frac{y}{r} = \frac{12}{13}$ Common misconceptions: - $-\frac{5}{13}$ is $\cos\theta = \frac{x}{r}$. - $-\frac{12}{5}$ is $\tan\theta = \frac{y}{x}$. - $\frac{12}{5}$ incorrectly uses $\frac{y}{|x|}$.

20 [Challenging] • Target: Limaçons and Special Polar Curves • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

The graph of the function $f(x) = A\sin(Bx + C) + D$ is shown below. The maximum value of the function is 5 and the minimum value is -1. The period of the function is $\frac{\pi}{2}$, and the graph passes through the point $(0, 2)$. Find the value of A.

Final Answer: 3. The amplitude A of a sinusoidal function is half the difference between its maximum and minimum values. \\ Given: Maximum value = 5, Minimum value = -1 \\ Amplitude x is 3. \\ The other parameters (period and phase shift) can be used to find B, C, and D, but are not necessary to determine A.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Coordinate Conversions (Polar <-> Rectangular)

Convert the polar point $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$ to rectangular coordinates (x, y).

a. $(-3\sqrt{3}, 3)$ ✓

b. $(3\sqrt{3}, -3)$

c. $(-3, 3\sqrt{3})$

d. $(3, -3\sqrt{3})$

$x = r\cos\theta = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. $y = r\sin\theta = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$. The rectangular point is $(-3\sqrt{3}, 3)$.

22 [Foundational] • Target: Coordinate Conversions (Polar <-> Rectangular)

Convert the rectangular point $(-4, 4)$ to polar coordinates with $r > 0$ and $0 \le \theta < 2\pi$.

a. $\left(4\sqrt{2}, \frac{3\pi}{4}\right)$ ✓

b. $\left(4\sqrt{2}, \frac{7\pi}{4}\right)$

c. $\left(8, \frac{3\pi}{4}\right)$

d. $\left(4\sqrt{2}, \frac{5\pi}{4}\right)$

$r = \sqrt{(-4)^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$. Since $(-4, 4)$ is in Quadrant II, $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.

23 [Foundational] • Target: Coordinate Conversions (Polar <-> Rectangular)

Which polar point represents the same location as $(r, \theta) = \left(3, \frac{\pi}{3}\right)$?

a. $\left(-3, \frac{4\pi}{3}\right)$ ✓

b. $\left(-3, \frac{\pi}{3}\right)$

c. $\left(3, -\frac{\pi}{3}\right)$

d. $\left(-3, \frac{2\pi}{3}\right)$

In polar coordinates, $(r, \theta) = (-r, \theta + \pi)$. For $\left(3, \frac{\pi}{3}\right)$, $\left(-3, \frac{\pi}{3} + \pi\right) = \left(-3, \frac{4\pi}{3}\right)$ represents the exact same physical point.

24 [Foundational] • Target: Converting Equations between Polar and Cartesian

Convert the polar equation $r = 6\cos\theta$ into rectangular coordinates.

a. $(x-3)^2 + y^2 = 9$ ✓

b. $x^2 + (y-3)^2 = 9$

c. $x^2 + y^2 = 6x$

d. $x^2 + y^2 = 36$

Multiply both sides by r: $r^2 = 6r\cos\theta$. Substitute $r^2 = x^2 + y^2$ and $x = r\cos\theta$: $x^2 + y^2 = 6x \Rightarrow x^2 - 6x + 9 + y^2 = 9 \Rightarrow (x-3)^2 + y^2 = 9$ (circle centered at $(3, 0)$ of radius 3).

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25 [Foundational] • Target: Converting Equations between Polar and Cartesian

Convert the rectangular equation $x^2 + y^2 = 16$ to polar form.

a. $r = 4$ ✓

b. $r = 16$

c. $r = 4\cos\theta$

d. $r = 16\sin\theta$

Since $r^2 = x^2 + y^2$, we have $r^2 = 16 \Rightarrow r = 4$.

26 [Foundational] • Target: Converting Equations between Polar and Cartesian

Convert $r = 4\csc\theta$ to rectangular coordinates.

a. $y = 4$ ✓

b. $x = 4$

c. $y = 4x$

d. $x^2 + y^2 = 16$

Multiply both sides by $\sin\theta$: $r\sin\theta = 4$. Since $y = r\sin\theta$, this is the horizontal line $y = 4$.

27 [Foundational] • Target: Rose Curves and Symmetry

How many petals does the rose curve $r = 5\cos(4\theta)$ have?

a. 8 ✓

b. 4

c. 16

d. 2

For a rose curve $r = a\cos(n\theta)$, if n is even, there are $2n$ petals. Here $n = 4$ is even, so the rose has $2(4) = 8$ petals.

28 [Foundational] • Target: Rose Curves and Symmetry

How many petals does the rose curve $r = 3\sin(5\theta)$ have?

a. 5 ✓

b. 10

c. 15

d. 3

If n is odd, the rose curve has exactly n petals. Since $n = 5$ is odd, the curve has 5 petals.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Rose Curves and Symmetry

What is the maximum distance from the pole (origin) to the curve $r = 4\sin(3\theta)$?

a. 4 ✓

b. 12

c. 3

d. 8

The maximum value of $|\sin(3\theta)|$ is 1. Therefore, the maximum radius is $|r| = 4(1) = 4$.

30 [Intermediate] • Target: Limaçons and Special Polar Curves

Classify the polar curve $r = 2 + 2\cos\theta$.

a. **Cardioid (heart-shaped limaçon)** ✓

b. Limaçon with an inner loop

c. Dimpled limaçon

d. Convex limaçon

For $r = a \pm b\cos\theta$, when $\frac{a}{b} = 1$, the curve is a cardioid.

31 [Intermediate] • Target: Limaçons and Special Polar Curves

Which polar equation produces a limaçon with an inner loop?

a. $r = 2 + 4\cos\theta$ ✓

b. $r = 4 + 2\cos\theta$

c. $r = 3 + 3\sin\theta$

d. $r = 5 + 4\cos\theta$

A limaçon $r = a + b\cos\theta$ has an inner loop when $\frac{a}{b} < 1$. For $r = 2 + 4\cos\theta$, $\frac{2}{4} = 0.5 < 1$.

32 [Intermediate] • Target: Limaçons and Special Polar Curves

At what angles $\theta \in [0, 2\pi)$ does the curve $r = 2 + 4\cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \frac{2\pi}{3}$ **and** $\theta = \frac{4\pi}{3}$ ✓

b. $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$

c. $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$

d. $\theta = \pi$

Set $r = 0 \Rightarrow 2 + 4\cos\theta = 0 \Rightarrow \cos\theta = -\frac{1}{2}$. In $[0, 2\pi)$, this occurs at $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$.

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33 [Intermediate] • Target: Coordinate Conversions (Polar <-> Rectangular)

Pythagorean identity: $1 + \tan^2\theta =$

a. $\csc^2\theta$

b. $\sec^2\theta$ ✓

c. $\cot^2\theta$

d. $\cos^2\theta$

Applying Unit 3 · Trigonometric and Polar Functions principles, $\sec^2\theta = 1 + \tan^2\theta$ Therefore, the verified correct option is $\sec^2\theta$.

34 [Intermediate] • Target: Converting Equations between Polar and Cartesian

Law of Cosines: $c^2 =$

a. $a^2 + b^2$

b. $a^2 + b^2 - 2ab\cos C$ ✓

c. $a^2 - b^2 + 2ab\cos C$

d. $2ab\cos C$

Applying Unit 3 · Trigonometric and Polar Functions principles, Standard Law of Cosines. Therefore, the verified correct option is $a^2 + b^2 - 2ab\cos C$.

35 [Intermediate] • Target: Rose Curves and Symmetry

$\sec x =$

a. $\frac{1}{\sin x}$

b. $\frac{1}{\cos x}$ ✓

c. $\frac{\sin x}{\cos x}$

d. $\frac{\cos x}{\sin x}$

Applying Unit 3 · Trigonometric and Polar Functions principles, Reciprocal identity. Therefore, the verified correct option is $\frac{1}{\cos x}$.

36 [Intermediate] • Target: Limaçons and Special Polar Curves

For $f(x) = 5\sin(1/2x - 2) + 2$, find the midline.

a. $y = 2$ ✓

b. $y = 3$

c. $y = 1$

d. $y = 5$

Applying Unit 3 · Trigonometric and Polar Functions principles, Midline $= y = D = y = 2.$ Therefore, the verified correct option is $y = 2$.

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37 [Intermediate] • Target: Coordinate Conversions (Polar <-> Rectangular)

For $f(x) = 4\sin(3x - 1) - 1$, find the amplitude.

a. **4** ✓ **b.** 8

c. 5 **d.** 3

Applying Unit 3 · Trigonometric and Polar Functions principles, Amplitude $= |A| = 4$. Therefore, the verified correct option is 4.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Converting Equations between Polar and Cartesian • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

For $f(x) = 3\sin(x - 1) - 2$, find the amplitude.

a. 2 **b.** 4

c. **3** ✓ **d.** 6

Applying Unit 3 · Trigonometric and Polar Functions principles, Amplitude $= |A| = 3$. Therefore, the verified correct option is 3.

39 [Challenging] • Target: Rose Curves and Symmetry • Cross-Lesson: 4.01 Inverse & Composite Functions

For $f(x) = 3\cos(2x + 2) - 4$, find the amplitude.

a. 4 **b.** 2

c. **3** ✓ **d.** 6

Applying Unit 3 · Trigonometric and Polar Functions principles, Amplitude $= |A| = 3$. Therefore, the verified correct option is 3.

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40 [Challenging] • Target: Limaçons and Special Polar Curves • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

For $f(x) = 3\sin(x + 2) - 3$, find the amplitude.

a. 6

b. **3** ✓

c. 4

d. 2

Applying Unit 3 · Trigonometric and Polar Functions principles, Amplitude $= |A| = 3$. Therefore, the verified correct option is 3.

41 [Challenging] • Target: Coordinate Conversions (Polar <-> Rectangular) • Cross-Lesson: 4.03 Parametric Equations & Modeling

Evaluate $\arcsin(\sqrt{3}/2)$.

a. $5\pi/6$

b. $\pi/3$ ✓

c. 0

d. $\pi/2$

$\arcsin(\sqrt{3}/2) = \pi/3$ since this angle lies in the range of $\arcsin$ and has the given trig value.

42 [Challenging] • Target: Converting Equations between Polar and Cartesian • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Evaluate $\arcsin(\sqrt{2}/2)$.

a. $3\pi/4$

b. π

c. $\pi/3$

d. $\pi/4$ ✓

$\arcsin(\sqrt{2}/2) = \pi/4$ since this angle lies in the range of $\arcsin$ and has the given trig value.

43 [Challenging] • Target: Rose Curves and Symmetry • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Evaluate $\arctan(\sqrt{3}/3)$.

a. 0

b. $\pi/2$

c. $\pi/6$ ✓

d. $2\pi/3$

$\arctan(\sqrt{3}/3) = \pi/6$ since this angle lies in the range of $\arctan$ and has the given trig value.

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44 [Challenging] • Target: Limaçons and Special Polar Curves • Cross-Lesson: 1.03 Rates of Change & Concavity

Evaluate $\arctan(\sqrt{3})$.

a. $5\pi/6$

b. $2\pi/3$

c. $\pi/2$

d. $\pi/3$ ✓

$\arctan(\sqrt{3}) = \pi/3$ since this angle lies in the range of $\arctan$ and has the given trig value.

45 [Challenging] • Target: Coordinate Conversions (Polar <-> Rectangular) • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Identify the polar curve $r = 5(1 + \cos\theta)$.

a. a circle

b. a rose with 4 petals

c. a limaçon with an inner loop

d. **a cardioid (heart-shaped curve)** ✓

The equation $r = 5(1 + \cos\theta)$ has the standard polar form matching: a cardioid (heart-shaped curve).

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